FlashSSM-3D: A Distilled State Space Model for Lightning-Fast Dense 3D Understanding

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Abstract

3D geometric understanding is undergoing a major transition where rigid, deterministic, optimization-based pipelines are being replaced by large-scale, deep learning models. While recent methods achieve impressive performance on tasks such as multi-view 3D reconstruction and camera pose estimation, they suffer from two critical limitations: **(i):** current state-of-the-art (SOTA) models require industry-level compute and training corpora, severely limiting accessibility, and **(ii):** recent approaches remain constrained to modifying existing pretrained transformer models, since substantially changing their architecture would require retraining from scratch. We instead study a complementary direction: training an attention-free state space model for dense 3D understanding. We introduce **FlashSSM-3D**, the first 3D reconstruction state space model capable of performing accurate geometric reasoning over hundreds of images with linear computational complexity. However, training such a model from scratch with competitive performance is impractical under typical academic compute constraints. To address this, we train FlashSSM-3D by distilling knowledge from VGGT, avoiding the need to learn 3D geometric priors entirely from scratch. However, standard distillation losses are poorly suited for dense 3D reconstruction, where preserving real-world spatial structure is essential. We therefore propose a **geometry-aware knowledge distillation** method specifically tailored for large-scale 3D reconstruction models. Our method introduces two novel losses: a Procrustes-based alignment loss that enforces global structural consistency between student and teacher 3D point predictions, and a camera reprojection loss that directly supervises view-consistent camera geometry between teacher and student. Extensive experiments across multi-view 3D reconstruction and camera pose estimation benchmarks demonstrate that FlashSSM-3D achieves competitive performance while requiring only $\sim 25\%$ of the training data and less than 10% of the training compute used by VGGT. At inference time, FlashSSM-3D benefits from the practical efficiency of state space models, reconstructing scenes from 500+ views in under 10 seconds while maintaining competitive accuracy.

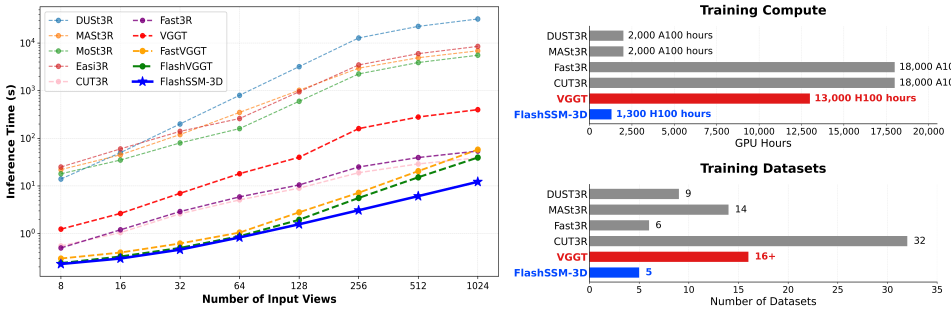


Figure 1: **FlashSSM-3D: a fast and training-efficient 3D reconstruction SSM.** FlashSSM-3D scales efficiently to hundreds of input views and requires *significantly* less compute and data to train than previous models.

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1 Introduction

3D geometric understanding lies at the core of many fundamental computer vision tasks, including autonomous navigation [1, 2], robotics [3, 4], and large-scale scene reconstruction [5]. Given one or more images (also referred to as views) as input, the goal of 3D geometric understanding is to produce structured geometric annotations of the scene - such as depth maps, 3D point clouds, and camera pose estimations - which jointly describe both scene structure and the spatial relationship between the camera and the environment. Classical 3D reconstruction and camera estimation methods were dominated by deterministic pipelines such as Structure-from-Motion (SfM) [5, 6] and Multi-View Stereo (MVS) [7, 8, 9]. However, they relied on handcrafted feature extraction, correspondence matching between paired images, and global alignment, limiting their flexibility and scalability [10, 11]. The rise of deep learning fundamentally reshaped this landscape, with early learning-based approaches introducing feed-forward networks trained on large-scale, image-paired datasets with 3D supervision. These models [12, 13, 14, 15] leverage dense matching and learned geometric priors to recover 3D structure from image pairs and were the first learnable, end-to-end methods to generally frame 3D reconstruction as a pointmap regression problem. While effective, these methods are typically limited to a fixed and small number of views, constraining their applicability in multi-view settings and often requiring expensive post-processing steps like bundle adjustment. More recently, a new class of models has emerged that can process an arbitrary number of input images and perform global 3D reasoning across views [16, 17, 18, 19, 20]. These newer models demonstrate impressive reconstruction quality by jointly reasoning over multiple frames using large transformer-based architectures, enabling scalable multi-view reconstruction and significantly improving robustness to viewpoint diversity and scene complexity.

Despite these advances, current feed-forward 3D models remain difficult to scale beyond the transformer architectures on which they are built. For example, VGGT [16] achieves strong performance using a 1B+ parameter transformer trained for over $13,000$ GPU hours, making substantial architectural changes infeasible without retraining at industry scale. As a result, recent acceleration methods [21, 22] alleviate inference cost by modifying existing transformer models, but they do not answer a key question: **can we move beyond attention-based 3D reconstruction models entirely, while maintaining competitive performance?**

To address these challenges, we propose **FlashSSM-3D**: a lightning-fast reconstruction model that demonstrates the efficacy of SSMs for feed-forward 3D reconstruction. FlashSSM-3D is a fully Mamba-based 3D reconstruction model that uses linear-time state-space layers for sequence mixing, enabling efficient inference on large-scale, multi-view scenes. However, moving from the attention mechanism to selective scanning for sequence modeling introduces a substantial representational gap, making it difficult to recover the global geometric reasoning of large transformer models through standard training alone. To bridge this gap, we train FlashSSM-3D by distilling knowledge from VGGT, avoiding the need to learn 3D geometric priors entirely from scratch. Yet, we find that the naive application of knowledge distillation is insufficient for fully transferring strong geometric understanding into an attention-free architecture, presumably due to the complexity of 3D understanding. Therefore, we introduce a novel **geometry-aware** knowledge distillation framework for 3D reconstruction. To the best of our knowledge, this is the first knowledge distillation approach explicitly designed for large-scale 3D reconstruction models. We propose two novel geometry-aware distillation losses: a Procrustes-based alignment loss that enforces global structural consistency between student and teacher 3D point predictions, and a camera reprojection loss that directly supervises view-consistent camera geometry between teacher and student. Together, these losses enable effective transfer of geometric reasoning into FlashSSM-3D, resulting in a fully Mamba-based 3D reconstruction model that benefits from the efficient long-sequence scaling of SSMs while maintaining competitive reconstruction and camera pose accuracy (Fig. 1). Our main contributions are as follows:

1. We introduce FlashSSM-3D, a Mamba-based 3D reconstruction model that validates the applicability of SSMs for multi-view 3D understanding.
2. We demonstrate that traditional knowledge distillation is insufficient for transferring strong 3D geometric priors across architectures, and propose a geometry-aware distillation framework tailored specifically for cross-architecture 3D understanding distillation.
3. We demonstrate that FlashSSM-3D achieves competitive multi-view reconstruction and camera pose estimation performance while requiring $\sim 25\%$ of the training data and $< 10\%$ training compute than VGGT.

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2 Related Works

2.1 State Space Models

Structured state space models (SSMs) [23, 24, 25] have recently gained significant attention as efficient sequence modeling architectures for vision applications, ranging from image classification [26] to video understanding [27]. Their appeal comes in part from their dual parameterization form: SSMs can be implemented as a recurrence for near-linear scaling with sequence length, or equivalently as a convolution for parallel GPU execution. Conceptually, SSMs map an input sequence to an output sequence by maintaining a learned latent state that evolves over time. Given an input signal $x(t) \in \mathbb{R}^L$, the hidden state $h(t) \in \mathbb{R}^N$ is updated according to a latent dynamical process and used to generate an output $y(t) \in \mathbb{R}^L$. This process is commonly formulated as a continuous-time linear system defined by parameters $(\mathbf{A}, \mathbf{B}, \mathbf{C})$: $h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t)$, and $y(t) = \mathbf{C}h(t)$, where $\mathbf{A} \in \mathbb{R}^{N\times N}$ controls the evolution of the latent state, while $\mathbf{B} \in \mathbb{R}^{1\times N}$ and $\mathbf{C} \in \mathbb{R}^{N\times 1}$ control input and output projections, respectively, of the hidden state. For discrete input sequences, these continuous dynamics are converted into a discrete system using a learnable step size Δ. Under a zero-order hold discretization, the parameters become:

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \qquad \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - I\right)\Delta\mathbf{B}, \qquad \bar{\mathbf{C}} = \mathbf{C}. \tag{1}$$

Thus, an SSM can be evaluated through two mathematically equivalent views. The recurrent form updates the hidden state sequentially (left), while the convolutional form applies a structured kernel constructed from repeated applications of the state transition matrix (right):

$$\begin{aligned} h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t, & \bar{\mathbf{K}} &= (\bar{\mathbf{C}}\bar{\mathbf{B}}, \bar{\mathbf{C}}\bar{\mathbf{A}}\bar{\mathbf{B}}, \cdots, \bar{\mathbf{C}}\bar{\mathbf{A}}^t\bar{\mathbf{B}}), \\ y_t &= \bar{\mathbf{C}}h_t, & y &= x * \bar{\mathbf{K}}. \end{aligned}$$

In practice, the convolutional formulation is often preferred during training due to its parallelizability, while the recurrent form is advantageous at inference time for efficient sequential evaluation with constant memory cost.

2.2 3D Reconstruction and Multi-View Geometry

A long-standing goal in 3D vision is to recover scene structure (e.g., depth, dense correspondences, and 3D point maps) and camera motion from one or more images. Classical pipelines rely on geometric optimization [5, 28, 29, 30, 31] but are often brittle under texture-less regions, dynamic content, or extreme viewpoint changes [32, 33]. Furthermore, they often suffer heavily from their reliance on iterative, post-processing techniques (e.g. bundle adjustment) to align their predictions. Recent feed-forward approaches instead learn to directly predict dense geometry from images, substantially improving robustness and enabling faster inference at scale.

Early progress in learned dense reconstruction, such as DUSt3R [12] and MASt3R [13], matched predicted dense 3D point maps between two views, enabling downstream alignment and reconstruction without classical feature matching. To address scalability beyond two views, recent methods predict globally consistent geometry from an arbitrary number of input images. VGGT [16] leverages larger amounts of data paired with dense prediction heads to learn multi-view aggregation for dense 3D reconstruction and camera estimation. Fast3R [18] similarly targets efficient multi-view reconstruction, emphasizing practical speed/accuracy trade-offs for dense 3D prediction over multiple images. Recent works have also explored [21, 22] improving inference efficiency through attention and token optimizations. Tangentially, some works focus on producing reconstructions from image streams, where images arrive sequentially and memory/compute must remain bounded [20, 19, 34]. While effective, these methods exclusively build upon existing pretrained transformer backbones, leaving unresolved whether fundamentally different sequence models can support feed-forward 3D reconstruction. This motivates our exploration of state space models; however, moving from transformers to SSMs introduces a challenging cross-architecture distillation problem, where the student must inherit geometric reasoning from an architecturally distinct teacher. To address this, we propose FlashSSM-3D: a large 3D reconstruction SSM trained with a geometry-aware distillation framework designed specifically for transferring 3D geometric priors across architectures.

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3 Methodology

FlashSSM-3D is an attention-less feed forward network capable of taking a set of images as input and producing camera pose and point map predictions. We start by providing some brief background and problem setup in Sec. 3.1, followed by architecture and training details in Sections 3.2 and 3.3, and discussion of our geometry-aware knowledge distillation framework in Sec. 3.4.

3.1 Background and notation

The problem of geometric understanding revolves around taking a set of unordered and unposed images $\mathbf{X} = \{x_i\}_{i=1}^N$, where $x_i \in \mathbb{R}^{H \times W \times 3}$ with height H and width W, and producing various 3D annotations as output. The objective of FlashSSM-3D is to take $\mathbf{X}$ as input and accurately produce the corresponding 3D annotations for a given scene: FlashSSM-3D: $\mathbf{X} \rightarrow (\mathbf{C}, \mathbf{P})$, where $\mathbf{C} = \{c_i\}_{i=1}^N$ is the set of all camera pose predictions for each image, and $\mathbf{P} = \{p_i\}_{i=1}^N$ is the set of all 3D point predictions for each pixel in each image. Specifically, we parameterize the camera predictions for a given image $c_i = [r_i, t_i, f_i,]$ following [16, 35], where the rotation quaternion $r_i \in \mathbb{R}^4$, translation vector $t_i \in \mathbb{R}^3$, and field of view $f_i \in \mathbb{R}^2$, respectively, are concatenated together. The predicted set of 3D points of an image $p_i \in \mathbb{R}^{H \times W \times 3}$ simply predicts the location of each pixel in the image in a 3D Euclidean space (x, y, z).

3.2 Model Architecture & Training

FlashSSM-3D is constructed with block-level correspondence to VGGT, allowing knowledge to be transferred from the transformer teacher into our attention-free student by replacing frame-wise and global self-attention with selective scan state-space layers. This substitution transforms the model from a quadratic-complexity transformer into a subquadratic state-space architecture, enabling scalable inference over hundreds of views. We use the same image encoder as VGGT (DINO [36]) to extract visual features for each image; yet, this has a negligible impact on runtime, which is dominated by the actual reconstruction section of the model (Fig. 2).

However, directly replacing attention with state-space layers introduces a substantial representational gap. While both architectures perform sequence mixing, attention relies on explicit pairwise token interactions, whereas selective scan performs recurrent updates in a learned latent space. As a result, training the student model from scratch would require prohibitively large-scale data and compute to recover VGGT's geometric reasoning capabilities, especially considering the resources used to train VGGT (16+ datasets, $13,000+$ GPU hours). To solve this, we leverage knowledge distillation to instill FlashSSM-3D with knowledge from VGGT. Crucially, because of the architectural mismatch between teacher and student and the complex nature of 3D geometric understanding, standard knowledge distillation does not fully suffice (see Sec. 4.5). Therefore, we adopt a multi-stage cross-architecture distillation strategy that progressively transfers geometric reasoning from the attention-based teacher to our SSM-based student.

3.3 Subquadratic Distillation

Our distillation process begins by aligning internal representations between teacher and student before imposing specific geometric supervision. We adopt the three-stage subquadratic distillation framework of [37], where the first two stages focus on representation-level alignment. We make significant changes to the vital third stage of distillation as discussed in Sec. 3.4.

3.3.1 Matrix Mixer Orientation.

The first stage of subquadratic distillation aims to align the student's *matrix mixer* mechanism with the teacher's *matrix mixer*. [37] defines a matrix mixer as the core sequence-mixing component of a block, such as the KQV matrices in attention or the state-evolution matrix $\mathbf{A}$ in SSMs. This stage aims to ensure that both student and teacher perform comparable token mixing within each block prior to downstream geometric supervision. Specifically, given $\mathbf{X}$ as input, both the teacher and student first map the images to hidden token embeddings through a shared image encoder: $\mathbf{H}^{(0)} = \mathrm{Enc}(\mathbf{X})$, where $\mathbf{H}^{(0)} \in \mathbb{R}^{B \times L \times d}$ denotes the initial token representations, B is the batch size, L is the sequence length, and d is the embedding dimension. The teacher consists of K stacked attention blocks $\mathcal{B}_T^{(k)}$ that produce corresponding hidden token representations $\mathbf{H}_T^{(k)}$ at each block:

$$\mathbf{H}_T^{(k)} = \mathcal{B}_T^{(k)}(\mathbf{H}_T^{(k-1)}), \quad k = 1, \ldots, K.$$

Let $\mathrm{TeacherMixer}(\cdot)$ denote the teacher's self-attention mixer and $\mathrm{StudentMixer}_{\boldsymbol{\phi}}(\cdot)$ denote the student's selective scan matrix mixer parameterized by $\boldsymbol{\phi} = \{\bar{\mathbf{A}}, \bar{\mathbf{B}}, \bar{\mathbf{C}}, \Delta\}$ (see Eq. 1). Within

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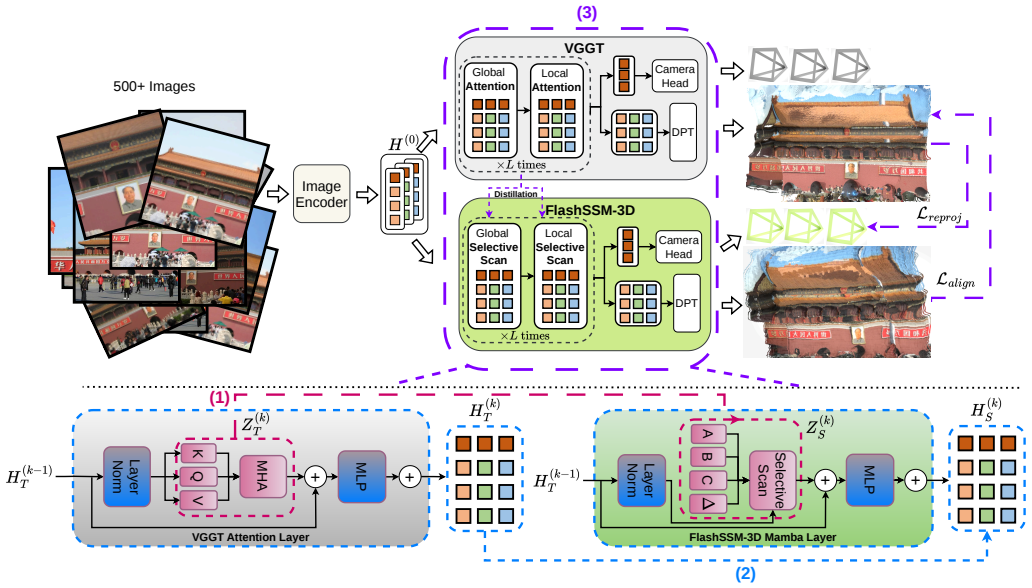


Figure 2: Distilling VGGT into FlashSSM-3D occurs over 3 stages: **(1): Matrix Mixer Orientation -** Aligns the student's internal selective scan output with the teacher's self-attention KQV output. **(2): Hidden-State Alignment -** Matches block-level representations between teacher and student by training all block components (e.g. linear layers) simultaneously. **(3): Geometry-Aware Knowledge Distillation -** Directly supervises 3D points and camera predictions using Procrustes alignment and reprojection losses. Collectively, this transfers VGGT's geometric reasoning into our student model.

each block, the mixer operates on the incoming hidden tokens to produce intermediate mixed token representations before the remaining block components (e.g., projections and residual connections) are applied. We denote the teacher's mixer output at the k-th block (left) and the student's mixer output (right) as:

$$\mathbf{Z}_T^{(k)} = \text{TeacherMixer}(\mathbf{H}_T^{(k-1)}), \qquad \mathbf{Z}_S^{(k)} = \text{StudentMixer}_{\boldsymbol{\phi}}(\mathbf{H}_T^{(k-1)}),$$

where $\mathbf{Z}_T^{(k)}, \mathbf{Z}_S^{(k)} \in \mathbb{R}^{B\times L\times d}$. To isolate the effect of only the student mixer during this stage (shown in pink in Fig. 2), the student mixer receives the same teacher hidden tokens as input. We freeze the teacher and optimize only the student mixer parameters $\boldsymbol{\phi} = \{\bar{\mathbf{A}}, \bar{\mathbf{B}}, \bar{\mathbf{C}}, \Delta\}$ by minimizing the Frobenius distance between corresponding matrix mixers across all blocks.

$$\min_{\boldsymbol{\phi}} \sum_{k=1}^{K} \left\| \mathbf{Z}_S^{(k)} - \mathbf{Z}_T^{(k)} \right\|_F^2. \tag{2}$$

This stage essentially ensures that the token representations produced by the student's matrix mixer at each block emulate those produced by the teacher, i.e., $\mathbf{Z}_S^{(k)} \approx \mathbf{Z}_T^{(k)}$. While this establishes compatibility at the core token interaction level despite the shift from attention to state-space recurrence, it does not guarantee equivalence at the block output level.

3.3.2 Hidden-State Alignment

Following matrix mixer orientation, the second stage of subquadratic distillation aligns the block-level outputs of the student and teacher models. While stage 1 ensures comparable intermediate block representations, differences may still arise in the remaining block components. This stage therefore aims to match the final hidden state from each student block to that of the corresponding teacher block (shown in blue in Fig. 2). Let $\boldsymbol{\theta}_S$ denote the full set of student block parameters, for example including linear projection and layer normalization weights. The student block parameters $\boldsymbol{\theta}_S$ are optimized by minimizing the ℓ_2 distance between corresponding block outputs across all layers:

$$\min_{\boldsymbol{\theta}_S} \sum_{k=1}^{K} \left\| \mathbf{H}_S^{(k)} - \mathbf{H}_T^{(k)} \right\|_2^2, \tag{3}$$

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where $H_S^{(k)}$ denotes the student block outputs computed via its own forward pass. Together, these two stages align the student and teacher at the intermediate feature level and can be performed in parallel across layers. The resulting objective encourages the student's Mamba blocks to mirror the output from the corresponding attention blocks in the teacher, providing a semi-aligned representation space to perform geometry-aware knowledge distillation. However, simple prediction alignment in standard knowledge distillation alone does not ensure geometric consistency (see Tab. 3). In 3D reconstruction, small discrepancies in intermediate features can accumulate across layers and manifest as large errors in global structure or camera pose, hence the need for stricter and more direct geometric supervision during distillation.

3.4 Geometry-Aware Knowledge Distillation

Let f_T and f_S denote the teacher and student networks, respectively. Recall that given a set of N input views $\mathbf{X} = \{x_i\}_{i=1}^N$, both models produce per-view predictions: $f_T(\mathbf{X}) = (\mathbf{C}^T, \mathbf{P}^T)$ and $f_S(\mathbf{X}) = (\mathbf{C}^S, \mathbf{P}^S)$, where $\mathbf{C} = \{c_i\}_{i=1}^N$ are camera pose predictions, and $\mathbf{P} = \{p_i\}_{i=1}^N$ are dense 3D point maps. Since VGGT produces dense geometric outputs rather than class probabilities, we apply direct regression losses between student and teacher predictions:

$$\mathcal{L}_{\text{KD}} = \sum_{i=1}^{N} \left(\|p_i^S - p_i^T\|_1 + \|c_i^S - c_i^T\|_1 \right), \tag{4}$$

where superscripts S and T denote student and teacher outputs, respectively. The difficulty with applying *only* standard knowledge distillation is that 3D reconstruction ultimately requires consistency in a real geometric space, which may extend beyond simple point-wise regression between student and teacher. To ensure that geometric reasoning itself is transferred to the student, we introduce two novel geometry-aware auxiliary losses to supervise the student on a deeper structural level.

3.4.1 Procrustes Alignment Loss

To enforce global structural consistency between student and teacher 3D reconstructions, we introduce a Procrustes alignment loss on the 3D points. Prior work has explored monocular scale ambiguity by performing *scale-only* alignment between predicted and reference geometries. However, scale alignment alone does not ensure that the predicted 3D points from the student model are aligned with the same rotation and translation as the teacher (see Sec. 4.5). Therefore, we align student and teacher reconstructions using a *full similarity transform*. Let $p_{i,j}^S \in \mathbb{R}^3$ and $p_{i,j}^T \in \mathbb{R}^3$ denote the 3D point at pixel j of view i predicted by the student and teacher, respectively. We compute the optimal scale $s^* \in \mathbb{R}$, rotation $\mathbf{R}^* \in SO(3)$, and translation $\mathbf{t}^* \in \mathbb{R}^3$ via the closed-form, differentiable Umeyama algorithm [38]:

$$(s^*, \mathbf{R}^*, \mathbf{t}^*) = \arg\min_{s,R,t} \sum_{i=1}^{N} \sum_{j=1}^{HW} \left\| sRp_{i,j}^S + t - p_{i,j}^T \right\|_2^2. \tag{5}$$

The Procrustes alignment loss is then defined as

$$\mathcal{L}_{\text{align}} = \sum_{i=1}^{N} \sum_{j=1}^{HW} \left\| s^*\mathbf{R}^* p_{i,j}^S + \mathbf{t}^* - p_{i,j}^T \right\|_2^2. \tag{6}$$

This loss is invariant to global scale and coordinate frame differences alike, while directly supervising the global 3D points predicted by the student. This enforces that the student's predicted geometry is globally consistent with the teacher's up to a known similarity transform.

3.4.2 Camera Reprojection Loss

While the Procrustes alignment loss enforces global consistency between student and teacher 3D reconstructions, it operates purely in a relative 3D coordinate frame and remains agnostic to how these points are observed from individual camera views. Consequently, globally aligned point clouds do not necessarily imply accurate camera pose estimation. In particular, errors in predicted rotations or translations may persist even when the reconstructed structure appears well aligned. To explicitly supervise the student's camera predictions, we introduce a *camera reprojection loss* that enforces view-consistent projections of the teacher's 3D points.

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Let $p_{i,j}^T \in \mathbb{R}^3$ denote the teacher-predicted 3D point corresponding to pixel j in view i, expressed in the teacher's world coordinate frame. Recall that camera predictions from both the student and teacher is parameterized as $\mathbf{c}_i = [r_i, t_i, f_i,] \in \mathrm{SO}(3) \times \mathbb{R}^3 \times \mathbb{R}^2$ and they share the same intrinsics matrix $\mathbf{K} \in \mathbb{R}^{3\times3}$. Under a pinhole camera model Π, a single 3D point $p_{i,j}$ is projected to image coordinates $\tilde{u} = \mathbf{K}\left(r_i * p_{i,j} + \mathbf{t}_i\right)$ and homogeneously normalized: $\tilde{\mathbf{u}} = \mathbf{K}\left(r_i * p_{i,j} + \mathbf{t}_i\right), \quad \Pi(p_{i,j}; \mathbf{c}_i) = \left(\frac{\tilde{u}_x}{\tilde{u}_z}, \frac{\tilde{u}_y}{\tilde{u}_z}\right)$. This operation first transforms the 3D point into the camera coordinate frame and then applies perspective division to obtain pixel coordinates. The intuition behind this is to use the teacher's 3D points and camera predictions as a fixed geometric reference, thereby isolating the student's camera pose errors from its 3D reconstruction quality. To specifically better align the student's camera predictions with respect to the teacher, we project the same fixed teacher 3D point $p_{i,j}^T$ using both the student-predicted camera parameters $\mathbf{c}_i^S$ and teacher camera parameters $\mathbf{c}_i^T$, and penalize their discrepancy in the image plane:

$$\mathcal{L}_{\text{reproj}} = \sum_{i,j} \left\| \Pi(p_{i,j}^T; \mathbf{c}_i^S) - \Pi(p_{i,j}^T; \mathbf{c}_i^T) \right\|_1 .$$

Unlike Procrustes alignment, which enforces global structural consistency in 3D space, the reprojection loss enforces view-level consistency under perspective geometry in the student's camera predictions with respect to the teacher. Therefore, the two geometry-aware losses are complementary: the camera reprojection loss primarily improves camera pose accuracy, while the Procrustes loss improves global 3D structure. Together, they enable geometry-consistent knowledge transfer without additional annotations or optimizations. Thus, the final Stage-3 training objective is:

$$\mathcal{L} = \lambda_{\text{KD}}\mathcal{L}_{\text{KD}} + \lambda_{\text{reproj}}\mathcal{L}_{\text{reproj}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}}.$$

4 Experiments

In this section, we evaluate the effectiveness of our proposed approach on both camera pose estimation and multi-view 3D reconstruction tasks. Importantly, we show **(1):** our geometry-aware knowledge distillation method effectively transfers VGGT's capabilities into FlashSSM-3D, **(2):** FlashSSM-3D achieves VGGT-level performance with less training data and compute than VGGT, and **(3):** FlashSSM-3D performs inference significantly faster than most models, highlighting the benefit of its linear complexity. We also provide training and testing details, as well as ablations on our knowledge distillation framework and number of inference views.

4.1 Training

FlashSSM-3D is trained on a much smaller subset of the datasets used to train VGGT. While VGGT was trained on $16+$ large-scale datasets spanning diverse indoor and outdoor environments, our model is trained using only five datasets: **MegaDepth** [39], **ScanNet** [40], **CO3Dv2** [41], **ARKitScenes** [42], and **BlendedMVS** [43]. These datasets collectively cover a wide range of scene types, camera motions, and geometric complexity. Furthermore, we only train FlashSSM-3D using 4 H100 GPUs for a total of $\sim 1,300$ GPU hours; a significant reduction in data and compute than the full VGGT training recipe (see Table 4). More implementation details can be found in the supplementary.

4.2 Evaluation Metrics

Following prior work [16, 18, 44, 45], we evaluate performance using standard metrics for both camera pose estimation and 3D reconstruction. For camera pose evaluation, we report *Relative Translation Error* (RTE) and *Relative Rotation Error* (RRE), which measure the angular error (in degrees) in translation and rotation, respectively, between the predicted and ground-truth camera. We also report AUC@30, which is the area under the accuracy–threshold curve up to 30° computed across various thresholds on RTE and RRE. For 3D reconstruction quality, we evaluate predicted geometry using error and completion metrics. Reconstruction error measures the average distance between predicted 3D points and the ground truth points, while reconstruction completion measures the average distance between the ground truth points and the predicted 3D points. Accuracy measures the percentage of ground truth surface points that are within a fixed distance threshold of the predicted points.

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4.3 3D Reconstruction Results

We first evaluate FlashSSM-3D on large-scale multi-view 3D reconstruction benchmarks to assess its ability to recover accurate scene geometry compared to VGGT. Following [46], we evaluate FlashSSM-3D on 5 different datasets in Table 1 to evaluate its reconstructions across indoor, object-centric, and scene-level settings: DTU [47] for object-centric reconstructions, and 7-Scenes [48], Neural-RGBD [49], ScanNet [40], and TUM-RGBD [50] for indoor reconstructions at both the object and scene level. For each scene, we randomly sample 10 input views unless otherwise specified. As shown in Table 1, FlashSSM-3D achieves performance comparable to VGGT across all reconstruction metrics, despite being trained with roughly 25% of the data and 10% of the compute. Notably, these results demonstrate that geometry-aware distillation enables FlashSSM-3D to faithfully inherit VGGT's implicit geometric reasoning, even under a fundamentally different state-space architecture. For example, transformer-based models that are specifically tailored for speed such as CUT3R [19] and Fast3R [18] suffer from suboptimal performance compared to VGGT in Table 1 due to the trade-off between model speed and accuracy. However, FlashSSM-3D combines the efficiency of attention-free sequence modeling with the strong geometric capabilities inherited through distillation from VGGT. This allows us to increase the number of views during inference, slightly improving reconstruction quality with a negligible increase in inference speed compared to prior works. Since DTU only has 49 views per-scene, it is omitted for extremely dense reconstruction.

Table 1: 3D reconstruction results across various indoor object-level, and scene-level datasets. Best results are in **bold**, with second best underlined. FlashSSM-3D outperforms all baselines in runtime due to its use of linear-time selective scans. While some models have competitive runtime, FlashSSM-3D significantly outperforms them in error due to VGGT knowledge distillation.

Method	TUM-RGBD		ScanNet		NRGBD		7-Scenes		DTU		Time
	Error ↓	Comp ↓	Error ↓	Comp ↓	Error ↓	Comp ↓	Error ↓	Comp ↓	Error ↓	Comp ↓	
Dense Views (50)											
DUSt3R [12]	0.620	0.474	1.19	0.940	0.046	0.030	0.032	0.041	1.284	1.349	$\sim 500s$
MASt3R [13]	0.209	0.211	0.890	0.770	0.043	0.024	0.025	0.039	1.374	1.409	$\sim 450s$
CUT3R [19]	0.740	0.595	0.794	0.752	0.346	0.184	0.179	0.097	4.710	2.413	$\sim 4s$
Fast3R [18]	0.711	0.337	0.725	0.777	0.128	0.067	0.045	0.047	3.647	2.319	$\sim 6s$
VGGT [16]	**0.105**	**0.101**	**0.434**	**0.411**	**0.015**	**0.012**	**0.019**	**0.032**	**1.125**	**1.295**	$\sim 9s$
FlashSSM-3D	0.124	0.140	0.466	0.423	0.028	0.024	0.020	0.036	1.22	1.38	**0.6s**
Extremely Dense Views (300+)											
DUSt3R [12]	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
MASt3R [13]	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
CUT3R [19]	0.690	0.651	0.808	0.757	0.132	0.056	0.041	0.029	-	-	$\sim 22s$
Fast3R [18]	0.698	0.322	0.733	0.727	0.061	0.031	0.040	0.056	-	-	$\sim 35s$
VGGT [16]	**0.074**	**0.083**	**0.433**	0.406	**0.016**	**0.017**	0.020	0.027	-	-	$\sim 130s$
FlashSSM-3D	0.100	0.092	0.458	**0.399**	0.043	0.026	0.028	0.034	-	-	**3.5s**

4.4 Camera Pose Estimation Results

Following [16], we further evaluate camera pose estimation accuracy on the RealEstate10K [51] and CO3Dv2 [41] datasets. As shown in Table 2, FlashSSM-3D matches or closely approaches VGGT's camera pose accuracy while operating in a fully attention-less manner. Importantly, both VGGT and FlashSSM-3D perform better when the number of inference views increases, however VGGT suffers from inefficient quadratic scaling, impacting inference speed. FlashSSM-3D is effectively able to ingest 500 frames of size 518×336 and still perform camera pose estimation either faster or quite comparably against previous works (CO3Dv2 does not have such dense views per scene).

4.5 Training Ablation

To better evaluate the impact of our novel losses on our knowledge distillation framework, we ablate both our Procrustes alignment and camera reprojection losses on the ScanNet and RealEstate10K datasets, respectively, in Table 3. Starting from only standard knowledge distillation, we observe moderate performance across both camera pose estimation and 3D reconstruction. Incorporating the camera reprojection loss yields a notable improvement in pose accuracy and slightly improves reconstruction quality, indicating that enforcing geometric consistency at the image formation level directly benefits camera estimation. In contrast, adding the Procrustes alignment loss primarily improves 3D reconstruction metrics, reducing reconstruction error and improving completeness. When both losses are combined, the model achieves the best overall performance, significantly reducing reconstruction error (0.458) and improving completeness (0.399).

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Table 2: Comparison on Rel10K (unseen) and CO3Dv2. Parentheses indicate how many frames are used during inference.

Methods	Rel10K	CO3Dv2	Time
	AUC@30↑	AUC@30↑	
Colmap+SPSG [11]	45.2	25.3	∼15s
PixSfM [52]	49.4	30.1	>20s
PoseDiff [45]	48.0	66.5	∼7s
DUSt3R [12]	67.7	76.7	∼7s
MASt3R [13]	81.8	76.4	∼9s
VGGSfM v2 [35]	78.9	83.4	∼10s
MV-DUSt3R [53]	71.3	69.5	∼0.6s
CUT3R [19]	75.3	82.8	∼0.6s
FLARE [14]	78.8	83.3	∼0.5s
Fast3R [18]	72.7	82.5	∼0.2s
FastVGGT [21]	84.3	86.5	<0.5s
FlashVGGT [22]	85.3	86.8	<0.5s
VGGT [16]	85.3	88.2	0.7s
FlashSSM-3D	83.3	86.8	**0.5s**
VGGT (300)	95.1	**88.5**	131.4s
FlashSSM-3D (300)	93.4	87.6	**3.5s**
VGGT (500)	OOM	OOM	OOM
FlashSSM-3D (500)	**95.2**	-	**5.9s**

Table 3: Ablation study showing the contribution of each loss component across camera pose estimation and 3D reconstruction on Rel10K and ScanNet, respectively.

Losses			Pose	3D Recon	
$\mathcal{L}_{KD}$	$\mathcal{L}_{reproj}$	$\mathcal{L}_{align}$	AUC↑	Err.↓	Comp.↓
✓	✗	✗	80.2	0.551	0.456
✓	✓	✗	82.6	0.545	0.428
✓	✗	✓	80.8	0.507	0.418
✓	✓	✓	83.3	**0.458**	**0.399**
Baseline VGGT			85.3	0.433	0.406

Table 4: Training scale and time for recent popular models. Our geometry-aware knowledge distillation enables FlashSSM-3D to emulate VGGT performance on TUM-RGBD with significantly less data and compute.

Model	Params	Training Time	Datasets	Error (↓)
DUSt3R [12]	∼ 0.6B	∼ 2,000 A100 hours	9	0.620
MASt3R [13]	∼ 0.7B	∼ 2,000 A100 hours	14	0.209
Fast3R [18]	∼ 0.6B	∼ 18,000 A100 hours	6	0.711
CUT3R [19]	∼ 0.8B	∼ 18,000† A100 hours	32	0.740
VGGT [16]	∼ 1B	∼ 13,000 H100 hours	16+	**0.105**
FlashSSM-3D	∼ 1B	∼ **1,300** H100 hours	**5**	0.124

† Hours not directly reported, estimated given open-source codebase

4.6 Ablating the Number of Inference Views

Table 5 ablates the number of frames used at inference time and runtime for camera pose estimation. While both VGGT and FlashSSM-3D benefit from more views during inference, VGGT severely suffers from a quadratically growing runtime, even with certain GPU-level attention optimizations in PyTorch. However, FlashSSM-3D benefits both from increasing performance with more views as well as small increases in runtime. This especially highlights the benefit of our geometry-aware knowledge distillation, as we are able to both leverage the capabilities of VGGT while simultaneously harnessing the efficiency of SSMs; an attribute not easily attained with regular distillation as we showed in the previous section.

Table 5: Camera pose estimation results with various number of views at inference. On Rel10K, using more frames significantly improves performance at the cost of runtime. FlashSSM-3D is capable of leveraging more frames while also keeping runtime low, while VGGT's runtime grows quadratically.

Model	# Images	Rel10K					CO3Dv2					Time
		AUC@30 ↑	RTE ↓	RRE ↓	acc@10 ↑	acc@20 ↑	AUC@30 ↑	RTE ↓	RRE ↓	acc@5 ↑	acc@10 ↑	
VGGT	10	85.3	1.08	3.99	81.4	91.1	88.2	1.06	3.9	63.7	**99.8**	0.7s
	20	89.4	**1.03**	2.64	89.4	97.1	**88.3**	1.01	**3.6**	**72.5**	**100**	1.9s
	50	93.8	1.07	1.54	96.7	98.6	**88.3**	**0.93**	**3.6**	71.6	**100**	8.0s
	100	94.6	1.06	**1.08**	**98.2**	99.1	87.1	0.99	3.7	71.6	**100**	27.7s
	200	**94.7**	1.06	**1.08**	**98.2**	**99.2**	86.1	1.06	3.9	65.6	**100**	101.9s
FlashSSM-3D	10	83.3	**0.97**	5.6	78.0	89.8	86.8	1.10	5.9	58.5	91.2	**0.2s**
	20	86.3	1.02	3.33	85.7	96.8	87.1	1.14	5.3	**70.1**	92.4	**0.3s**
	50	90.3	0.99	1.79	94.1	97.6	87.2	1.08	4.3	66.2	93.0	**0.6s**
	100	92.4	1.00	1.50	95.6	97.2	**87.3**	**1.02**	**4.2**	67.6	**93.1**	**1.2s**
	200	**93.4**	1.00	**1.16**	**96.6**	**98.2**	87.1	**1.02**	**4.2**	67.6	**93.1**	**2.7s**

5 Conclusion

We introduce FlashSSM-3D, the first fully attention-less, feed-forward model for large-scale 3D reconstruction and camera pose estimation. Departing from existing pretrained transformer backbones, FlashSSM-3D explores whether SSMs can serve as a viable alternative architecture for dense multi-view geometric reasoning. By replacing self-attention with selective scan state-space layers, FlashSSM-3D achieves near-linear scaling complexity with respect to the number of input views while preserving strong geometric reasoning. However, this architectural shift introduces a challenging cross-architecture distillation problem. To address this, we propose a geometry-aware distillation framework combining Procrustes alignment and camera reprojection losses, enabling effective transfer of structural and camera pose consistency to a student model. Extensive experiments demonstrate that FlashSSM-3D achieves competitive performance while requiring significantly less data and compute, with substantial speedups when scaling to hundreds of views. These results suggest that attention-free, state-space architectures can provide a scalable path toward accurate feed-forward 3D reconstruction.

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A Appendix

A.1 Limitations

While FlashSSM-3D demonstrates that state space models can serve as effective backbones for feed-forward 3D reconstruction, several limitations remain. First, current deep learning frameworks and GPU kernels are still far more optimized for transformer attention than for selective scan operations, making SSM performance more sensitive to implementation details and backend support. Second, our model relies on VGGT as a teacher, meaning the quality of the distilled student is ultimately bounded by the geometric priors and failure modes of the teacher. Third, although FlashSSM-3D substantially improves inference speed, its memory usage remains comparable to recent transformer-acceleration methods rather than consistently lower. Further GPU optimizations to drop peak memory usage below transformer models remains an important direction for future work.

A.2 Architecture Details

FlashSSM-3D follows the same overall backbone structure as VGGT, differing only in the sequence mixing mechanism. We retain the same image encoder and tokenization pipeline as VGGT, where each input image is encoded using a pretrained DINO backbone to produce visual tokens. In VGGT, these tokens are processed by a stack of 24 transformer-style blocks, each with one frame-wise self-attention layer and one global self-attention, resulting in 48 layers. Following the subquadratic distillation framework of [37], we replace each self-attention layer in every VGGT block with a Mamba layer while keeping the remaining block components (e.g., normalization layers, linear projections, and residual connections) unchanged. This results in a one-to-one mapping between teacher and student blocks, allowing the student to mirror the representational structure of VGGT while replacing quadratic attention with linear-time state-space sequence mixing. All prediction heads (camera and pointmap heads) and other architectural details remain unchanged from VGGT.

A.3 Qualitative Results

We provide qualitative comparisons on unseen scenes in Figures A1 and A2. FlashSSM-3D produces dense 3D geometry that is visually comparable to VGGT, at a significantly lower runtime. For 100+ input views, FlashSSM-3D reconstructs the scene nearly $25\times$ faster than VGGT, highlighting the practical benefit of replacing quadratic attention with linear-time selective scan layers. Visually,

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FlashSSM-3D preserves fine geometric structure and completeness across complex architectural regions, which are common failure cases for fast reconstruction models. These results further demonstrate that, despite the architectural shift from attention to selective scanning with state space modeling, our geometry-aware distillation framework effectively transfers the teacher's geometric understanding while enabling substantially faster inference.

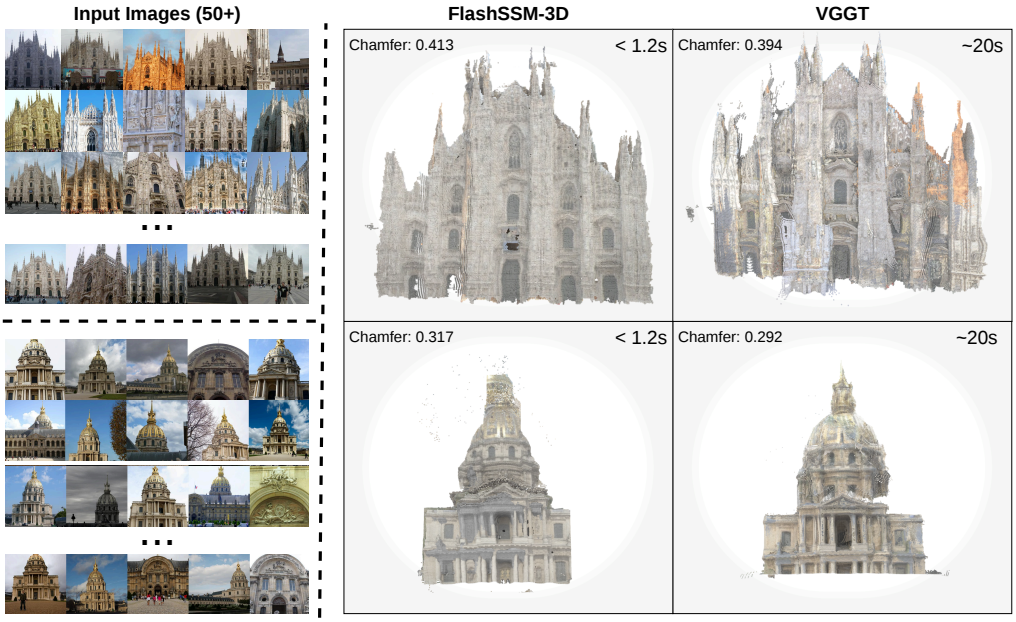


Figure A1: Qualitative results on **unseen** scenes. FlashSSM-3D retains VGGT-level accuracy (shown using Chamfer distance) while being 16x faster when reconstructing a dense number of views.

A.4 Computational Cost with Contemporary Work

To further evaluate inference scalability, we compare FlashSSM-3D against very recent VGGT acceleration methods released contemporaneously with this work in Table A1. FastVGGT and FlashVGGT reduce the cost of VGGT through token and attention optimizations, respectively, while FlashSSM-3D departs from attention entirely by using state-space sequence mixing. Across all tested input sizes, FlashSSM-3D achieves the lowest wall-clock runtime, with the gap increasing as the number of views grows. For example, at 1000 input images, FlashSSM-3D runs in 11.93s compared to 38.10s for FlashVGGT and 79.31s for FastVGGT. FlashSSM-3D does not achieve the lowest peak memory, but its memory usage remains comparable to these optimized transformer baselines. These results support our main claim that geometry-aware distillation enables an attention-free SSM backbone that is not only accurate, but also practically efficient for dense multi-view reconstruction.

Table A1: Comparison of runtime and GPU memory consumption across different numbers of input images. Best results are in bold.

Metric	Model	200	400	600	800	1000
Time (s) ↓	FastVGGT [21]	6.45	16.63	32.19	52.01	79.31
	FlashVGGT [22]	4.05	9.84	17.25	26.44	38.10
	FlashSSM-3D	**2.76**	**4.80**	**7.17**	**9.55**	**11.93**
GPU Mem. (GB) ↓	FastVGGT [21]	19.34	32.66	45.97	59.29	72.60
	FlashVGGT [22]	**16.97**	**27.92**	**38.83**	**49.76**	**60.68**
	FlashSSM-3D	21.80	36.63	50.95	65.26	79.57

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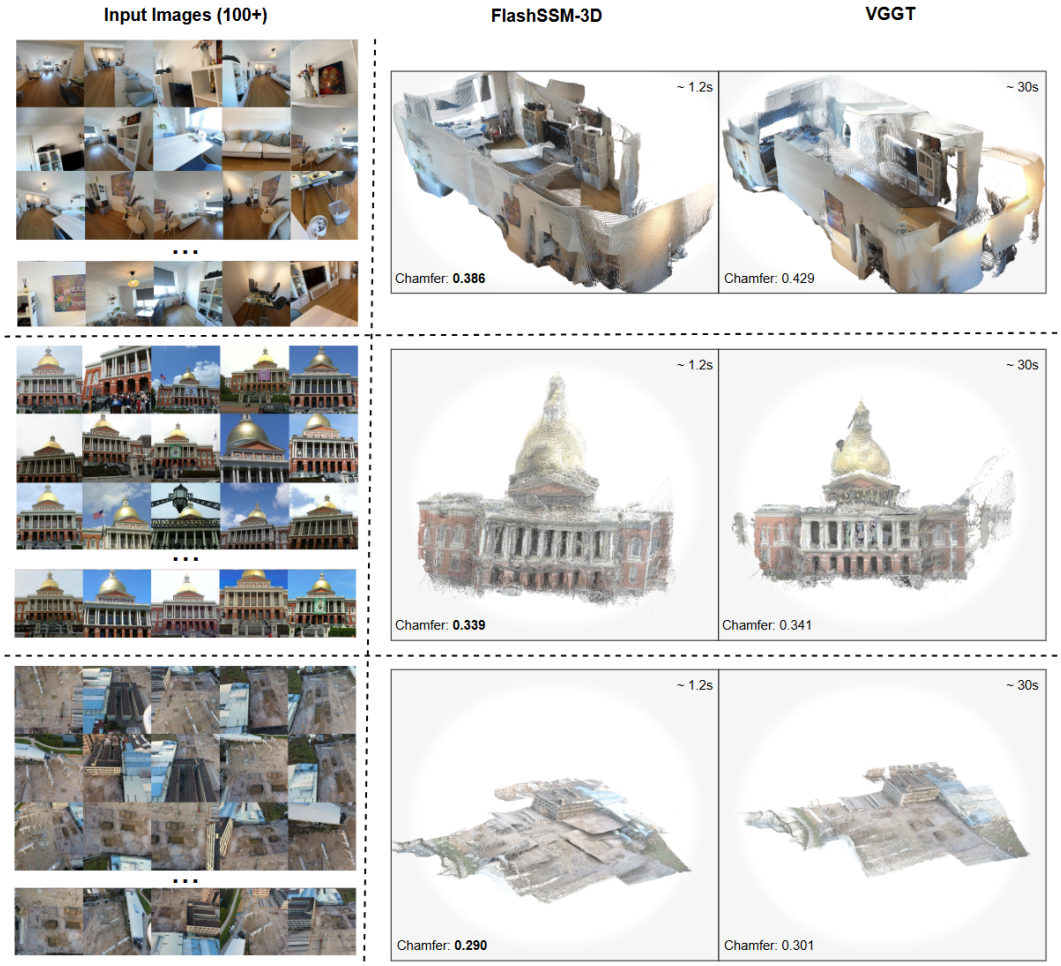


Figure A2: **Best viewed with zoom.** More qualitative results comparing VGGT and FlashSSM-3D on a large number of unseen images. In some cases, FlashSSM-3D performs better (lower Chamfer Distance) than VGGT with more coverage across the entire scene. This demonstrates that FlashSSM-3D has strongly recovered VGGT's 3D geometric understanding across diverse dense reconstruction scenarios, such as outdoor scenes and indoor scenes with many objects.

A.5 Training Settings

We provide additional details regarding the training configuration in Table A2. All stages are optimized using the AdamW optimizer with gradient clipping set to 1.0 and gradient accumulation of 4 steps. Training proceeds in three stages with progressively smaller learning rates and longer training schedules. The effective batch size is kept consistent during the first two stages and slightly reduced in the final stage to accommodate a larger number of input views. Across stages, the number of views used during training is gradually increased, enabling the model to first learn stable representations from small view sets before scaling to more complex multi-view reasoning. During Stage 3, we additionally perform a short warmup where training begins at a resolution of 224×224 for approximately 5% of the training steps before switching to the full 518×336 resolution for the remainder of training. All experiments are conducted on NVIDIA H100 GPUs with 80GB memory. Training is implemented in PyTorch using distributed training with the NCCL backend. The student network is wrapped using Fully Sharded Data Parallel (FSDP) with the `FULL_SHARD` strategy to reduce GPU memory consumption.

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Table A2: Optimization settings used in each training stage.

Setting	Stage 1	Stage 2	Stage 3
Optimizer	AdamW	AdamW	AdamW
Learning Rate	5×10^{-4}	2×10^{-4}	1×10^{-4}
Gradient Accumulation	4	4	4
Effective Batch Size	16	16	8
Number of Views	2-4	2-8	2-16
Gradient Clipping	1.0	1.0	1.0
Training Steps	50k	100k	200k
Image Resolution	224×224	224×224	224×224 / 518×336

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